Body parts used

- **Body curvature:** all **consecutive triples** of body parts in your bp_names order (e.g., bp1–bp2–bp3, then bp2–bp3–bp4, ...).
- **Tail-beat features:** the "tail extras" group: **bodypart6–bodypart11** (whatever of these exist in your project).
- **Inter-body coordination:** head group **(bodypart1, bodypart2, bodypart13)** vs tail group **(bodypart15, bodypart16)**.
- **Lateral symmetry: all** body parts' **x** coordinates.

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1) Body Curvature (how bent the fish is)

Columns

- **Segment_angle_{bp1}_{bp2}_{bp3}** — Angle at the **middle** point bp2 formed by the vectors bp1→bp2 and bp3→bp2.
  Units: **degrees** (signed; direction of bend sets the sign).
- **Total_body_curvature** — Sum of all segment angles in the frame (a whole-body bend score).
  Units: **degrees** (larger magnitude = more overall bend).
- **Body_curvature_change** — Frame-to-frame change of Total_body_curvature.
  Units: **degrees/frame**.
- **Body_curvature_mean_{W}**, **Body_curvature_std_{W}** — Rolling mean / std of Total_body_curvature over the last **W frames**.
  Units: **degrees** / **degrees**.

Body parts used: consecutive triplets along your skeleton order from bp_names.csv.

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2) Tail-Beat Features (frequency, amplitude, spectrum)

We average the **movement** (mm/frame) across the **tail extra** parts and analyze that signal.

Columns

- **Tail_movement_avg** — Mean of {bp}_movement across tail extras this frame.
  Units: **mm/frame**.

- **Tail_beat_frequency** — Peaks-per-second in Tail_movement_avg.
  Units: **Hz** (beats per second).
- **Tail_beat_amplitude** — Mean(peak) − Mean(trough) of Tail_movement_avg.
  Units: **mm/frame** (it's the amplitude of a movement-per-frame signal).
- **Tail_dominant_freq**, **Tail_dominant_freq_power** — Top frequency from Welch PSD and its power.
  Units: **Hz** / **power (arbitrary PSD units)**.
- **Tail_secondary_freq**, **Tail_secondary_freq_power** — Second strongest spectral component.
  Units: **Hz** / **power**.
- **Tail_spectral_entropy** — Entropy of the normalized PSD.
  Units: **bits** (via log2).

Body parts used: any of **bodypart6–bodypart11** that exist in your project.

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3) Inter-Body Coordination (head vs tail synchrony)

We compare average head movement to average tail movement.

Columns

- **Head_movement_avg** — Mean of {bp}_movement over head parts.
  Units: **mm/frame**.
- **Tail_movement_avg_for_coord** — Mean of {bp}_movement over tail parts.
  Units: **mm/frame**.
- **Head_Tail_corr** — Rolling Pearson correlation between the two signals (window ≈ **fps/2** frames).
  Range: **−1…+1** (−1 = perfectly out of phase, +1 = perfectly in phase).

Body parts used: head **(bodypart1, bodypart2, bodypart13)** and tail **(bodypart15, bodypart16)** (only those that exist).

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4) Lateral Symmetry (left–right spread in x)

Columns

- **Lateral_width** — Framewise **max(x) − min(x)** across all body parts.
  Units: **pixels** (current code uses raw coordinates).
- **Lateral_width_std** — Rolling std of Lateral_width (window ≈ **fps/2** frames).
  Units: **pixels**.

Body parts used: all body parts' **x** coordinates.